Monitoring tools · Quick concepts

Interview revision sheet. Read it the evening before, then test yourself in Interview practice.

Observability basics

- Three pillars: metrics (numbers over time), logs (events), traces (request path).
- Golden signals: latency, traffic, errors, saturation.
- RED for services: Rate, Errors, Duration. USE for resources: Utilization, Saturation, Errors.
- Use p95 and p99 latency, not just averages.

SLI, SLO, SLA

- SLI: the measured value, such as success rate.
- SLO: the target, such as 99.9%. SLA: the customer contract, often with penalties.
- Error budget: 100% minus the SLO; burn-rate alerts warn when it is used too fast.

Alerting

- Alert on symptoms users feel (error rate, latency), not every cause.
- Avoid alert fatigue: tune thresholds, use "for" durations and group alerts.
- Every alert should have an owner and a runbook.

Prometheus and Grafana

- Prometheus scrapes /metrics endpoints; query with PromQL, such as rate(http_requests_total[5m]).
- Counter only goes up; gauge goes up and down; histogram gives buckets for percentiles.
- Avoid high-cardinality labels such as user_id.
- Alertmanager routes alerts; Grafana draws dashboards from many sources.

ELK and Azure Monitor

- Elasticsearch stores, Logstash and Beats ship and parse, Kibana visualises.
- Azure Monitor: metrics, Log Analytics (KQL), Application Insights (APM), action groups for alerts.
- KQL example: Requests, then where Success == false, then summarize count() by bin(TimeGenerated, 5m).

Tracing and logging

- Span = one timed operation; trace = tree of spans for a request.
- OpenTelemetry is the vendor-neutral standard for traces, metrics and logs.
- Structured JSON logs with correlation IDs make searching easy.
- Mask sensitive data (passwords, card numbers) in logs.

Linux quick checks

- top or htop (CPU, memory), df -h (disk), free -h (memory), tail -f (follow logs).

Kubernetes probes

- Liveness failure restarts the container; readiness failure removes it from traffic.
- Spring Boot Actuator plus Micrometer expose JVM and app metrics.